

Business Intelligence-Driven Analysis of Hospital Encounter Patterns and 30-Day Readmission Using Power BI: A Case Study on Synthetic Patient Data

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Abstract

Hospital operations generate large volumes of encounter-level data that, if effectively analyzed, can inform readmission management, resource allocation, and payer coverage planning. This paper presents a business intelligence (BI) approach to hospital patient analysis using Microsoft Power BI, applied to a synthetic patient dataset generated using the Synthea patient population simulator, comprising 974 patients and 27,891 encounters recorded between 2011 and 2022. A relational data model was constructed linking patient demographic attributes to individual encounter records, and an interactive dashboard was developed incorporating key performance indicator (KPI) cards, trend visualizations, and Data Analysis Expressions (DAX)-based measures. Encounter-level analysis shows that ambulatory encounters accounted for the largest share of visits (44.95%), while inpatient encounters, though comprising only 4.07% of total encounters, recorded the longest average duration (36.84 hours) and highest average payer coverage (3,250.01 USD) among encounter classes. A 30-day readmission analysis of inpatient encounters found a patient-level readmission rate of 26.80%, with encounter-level readmission incidence higher among female patients (52.54%) than male patients (42.12%). These findings demonstrate the capacity of a structured BI dashboard to surface operationally significant patterns from large-scale encounter data, and the paper discusses the implications, scope, and limitations of using synthetic population data for this type of analysis.

Keywords: healthcare business intelligence, Power BI, hospital readmission, DAX, data visualization, synthetic patient data, Synthea

I. Introduction

The digitization of hospital operations has produced large volumes of structured data describing patient demographics, encounter history, and billing outcomes. Effective use of this data can support hospital administrators in managing readmission risk, forecasting resource demand, and understanding payer coverage patterns. However, translating raw encounter-level records into actionable insight requires both a well-structured data model and an accessible analytical interface.

Business intelligence (BI) platforms such as Power BI address this need by enabling interactive dashboards that combine descriptive statistics, trend analysis, and custom calculated measures without requiring end users to write code.

This paper presents a case study in applying Power BI to hospital encounter data at a substantial scale: 974 patients and 27,891 encounters. Because real hospital records are subject to strict privacy regulation and are not readily accessible for open academic work, this study uses a synthetic patient population generated by Synthea, an open-source patient population simulation tool developed for research and software testing purposes. Synthea-generated data preserves realistic statistical relationships between demographic attributes, encounter types, and clinical outcomes while containing no real individual's protected health information, making it suitable for methodological demonstration and dashboard validation.

The remainder of this paper is organized as follows. Section II reviews related work in healthcare business intelligence. Section III describes the dataset and methodology. Section IV describes the dashboard design. Section V presents the analytical results. Section VI discusses the findings and limitations. Section VII concludes the paper and outlines future work.

II. Related Work

Business intelligence and analytics have been widely studied as tools for improving healthcare decision-making. Prior work has examined how big data analytics capabilities can be operationalized within healthcare organizations to support both clinical and administrative outcomes. Other studies have surveyed the promise and challenges of applying big data analytics broadly across the healthcare sector, noting that data quality, integration, and interpretability remain persistent obstacles. Readmission prediction has received particular attention in the clinical informatics literature, with systematic reviews evaluating the performance of statistical and machine learning models for forecasting hospital readmission risk. The present study differs from this body of work by focusing specifically on the design and descriptive analytical output of a self-service BI dashboard, rather than on predictive model development, and by using an openly available synthetic dataset to allow full methodological transparency and reproducibility.

III. Dataset and Methodology

A. Data Source

The dataset used in this study was generated using Synthea, an open-source synthetic patient generator that produces realistic but fictitious patient records, including demographics, encounters,

procedures, and payer information, based on publicly available clinical and statistical models. The dataset comprises five primary tables: patient (974 records, 24 attributes including gender, marital status, age category, and county), encounters (27,891 records, 15 attributes including encounter class, start and stop timestamps, payer coverage, and total claim cost), procedures (47,701 records), payers (10 records), and organizations (1 record). Because the data is synthetic, it contains no real patient information and is not subject to the ethical or regulatory restrictions applicable to real clinical records, while still preserving realistic statistical structure suitable for demonstrating BI methodology.

B. Data Model

A relational data model was constructed in Power BI, establishing a one-to-many relationship between the patient table and the encounters table using patient identifier as the linking key. This model allows encounter-level metrics to be filtered and aggregated dynamically by any patient demographic attribute, including gender, age category, marital status, and county of residence.

C. Measures and Calculations

Custom DAX measures were defined for total patient count, average encounter duration (computed from encounter start and stop timestamps), average payer coverage, and age-category classification. Because the raw dataset does not include an explicit readmission flag, a 30-day readmission indicator was derived for this study: an inpatient encounter was classified as a readmission if its start date fell within 30 days of the same patient's immediately preceding inpatient encounter's discharge date. This definition follows a commonly used convention in hospital readmission analysis.

IV. Dashboard Design

The dashboard incorporates four primary components. KPI cards present headline metrics, including total patient count, overall encounter volume, average encounter duration, and gender distribution. A combination line-and-column chart displays yearly encounter volume and average duration segmented by age category, enabling trend analysis across the 2011-2022 study period. Donut charts visualize the proportional breakdown of patient gender and marital status. A detail table presents encounter class, patient identifier, encounter duration, and payer coverage for granular inspection. Interactive slicers for gender and county allow dashboard users to filter all visuals dynamically without modifying the underlying report structure.

V. Results

A. Patient Demographics

The patient population comprised 974 individuals, with a near-even gender split of 50.72% male and 49.28% female. By age category, the population skewed heavily toward elderly patients, with 85.32% classified as ELDER and 14.68% as SENIOR CITIZEN, reflecting the demographic focus of the underlying synthetic population sample. Marital status distribution showed 80.49% of patients recorded as married and 19.40% as single. Patients were distributed across five counties, with Suffolk County accounting for the largest share (644 of 974 patients).

B. Encounter Class Distribution

Across the 27,891 recorded encounters, ambulatory visits were the most frequent encounter class (44.95%), followed by outpatient (22.59%), urgent care (13.14%), emergency (8.33%), wellness (6.92%), and inpatient (4.07%) encounters.

Encounter Class	Share of Encounters (%)	Avg. Duration (hrs)	Avg. Payer Coverage (USD)
Ambulatory	44.95	9.48	1,029.23
Outpatient	22.59	5.88	599.64
Urgent Care	13.14	0.25	834.40
Emergency	8.33	1.54	1,398.29
Wellness	6.92	0.25	2,289.88
Inpatient	4.07	36.84	3,250.01

Although inpatient encounters represented the smallest share of total visits, they were associated with substantially longer average duration and higher average payer coverage than any other encounter class, consistent with the higher clinical intensity typically associated with inpatient admission.

C. Readmission Analysis

Of the 1,135 total inpatient encounters, 153 unique patients accounted for at least one inpatient admission. Applying the 30-day readmission definition described in Section III, 41 of these 153 patients (26.80%) experienced at least one readmission within 30 days of a prior inpatient discharge. At the encounter level, 552 of 1,135 inpatient encounters (48.63%) met the 30-day readmission criterion, indicating that a subset of patients accounted for a disproportionate share of repeat inpatient admissions.

Demographic Group	Encounter-Level 30-Day Readmission Rate
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	(%)
Female patients	52.54
Male patients	42.12
ELDER category	52.14
SENIOR CITIZEN category	45.22

D. Temporal Trends

Encounter volume varied across the 2011-2022 study period, ranging from 221 encounters in the partial final year (2022) to a peak of 3,883 encounters in 2014, with overall volume remaining in the range of approximately 2,200 to 3,500 encounters per year across most of the study period. This trend visualization allows dashboard users to identify periods of elevated encounter volume that may correspond to operational or population-level factors.

VI. Discussion

The results illustrate how a structured BI dashboard can surface clinically and operationally relevant patterns from encounter-level data at meaningful scale. The concentration of readmissions among a relatively small subset of patients (41 of 153 inpatient patients accounting for 552 readmission-flagged encounters) is consistent with findings in the broader readmission literature, where a minority of high-utilization patients often account for a disproportionate share of repeat hospitalizations. The observed difference in readmission incidence between female and male patients, and between ELDER and SENIOR CITIZEN age categories, suggests that demographic segmentation within a BI dashboard can help hospital administrators target readmission-reduction interventions more precisely.

A. Limitations

Several limitations should be considered when interpreting these results. First, the underlying dataset is synthetic, generated using the Synthea simulation framework rather than drawn from real hospital records; while Synthea is designed to preserve realistic statistical relationships, the specific readmission and demographic patterns reported here should not be interpreted as clinical findings about real patient populations. Second, the 30-day readmission definition applied in this study is a methodological approximation based on inpatient encounter sequencing, and does not account for planned readmissions, transfers, or clinical diagnosis codes that a full clinical readmission study would typically incorporate. Third, the patient population in this dataset is concentrated among elderly age categories, which may limit the generalizability of demographic comparisons to broader hospital populations. Future application of this dashboard methodology to

real, ethically sourced hospital data would be necessary to validate these findings for clinical or operational decision-making purposes.

VII. Conclusion and Future Work

This paper presented a Power BI-based business intelligence dashboard for analyzing hospital encounter data, applied to a large-scale synthetic patient dataset comprising 974 patients and 27,891 encounters. The dashboard's data model, DAX-based measures, and visual design were described, and analysis revealed measurable patterns in encounter class distribution, duration, payer coverage, and 30-day readmission incidence across demographic groups. These findings demonstrate the practical value of self-service BI tools for converting large volumes of structured healthcare data into actionable insight. Future work includes validating the dashboard methodology against real, properly anonymized hospital data under appropriate ethical oversight, incorporating clinical diagnosis and comorbidity data to refine the readmission definition, and extending the dashboard with predictive readmission risk scoring using machine learning models integrated with the existing Power BI data model.

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